



FIRST SEMESTER 2026-2027
COURSE HANDOUT

Dated: 1st August, 2026

Course No. : BIO G523
Course Title : Advanced & Applied Microbiology
Instructor In-charge : Prabhat Nath Jha (Email: prabhatjha@pilani.bits-pilani.ac.in; Room: 3277-V)
Instructor : Sandhya Marathe
Lab instructor : Juhi Mishra, Jennifer

1. Course of Description: Molecular taxonomy, Systematic Microbiology, Study of molecular diversity of microorganisms, Molecular tools employed in the study of microbial ecology, clinical microbiology, human-microbe interaction, molecular plant-microbe interaction, applied microbiology, and synthetic microbiology.

2. Scope & Objective of the Course:

This course deals with an in-depth study of microbial taxonomy and evolution, as well as the molecular aspects of microbe-host interactions. In addition, it includes applied aspects of microorganisms being utilized in industry and human health. It also emphasizes on recent development in microbial genomics, nanotechnology and biotechnology.

3. Text Book (TB):

Madigan M.T., Martinko J.M., Bender K.S., Buckley D.H., Stahl D.A. (2017). Brock Biology of Microorganism, 14th Ed., Pearson International Education

4. Reference Book (RB):

1. Wiley, J.M., Sherwood, L.M., Woolverton, C.J. (2007). Prescott, Harley, and Klein's Microbiology, 7th Ed. McGraw-Hill International Edition.
2. Glazer, A.N. and Nikaido, H. (2008). Microbial Biotechnology, Fundamentals of Applied Microbiology, 2nd Ed., Cambridge.

5. Course Plan:

Module No.	Lecture session	Ref. to Chapters	Learning Outcome
1. Introduction	L1: Introduction to the course		Getting introduced to the course
2. Microbial systematics	L 2- 4: Chemical and molecular methods for identification, species concept, molecular phylogeny, and typing of strains	TB-12, RB1-19	Understanding the process of evolution and methods to identify bacterial strains
3. Microbial Ecology	L 5-9: Microbial ecology, ecosystem services, surface and microbial assemblages, Ecology of soil, and marine ecosystem. Host-associated microbiome (including human)	TB-19, 22	Understanding of microbes with respect to the environment
	L 10 – 11: Methods in microbial ecology, Culture-dependent and independent analysis of microbial communities, metagenomics, stable isotope probing, Application of metagenomics in bioprospecting of drugs and enzymes.	TB-18. Reviews	Understanding of tools and techniques used to study microbial ecology



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

4. Plant Microbiology	L12 – 14: Plant growth promoting microorganisms; Associative bacteria, Endophytic bacteria: mechanisms of colonization; Biocontrol; Mycorrhiza	RB-29 TB-22	Identifying the role and importance of microbes in sustainable agriculture
	L15 – 16: Molecular basis of legume-rhizobia interaction, plant-pathogenic bacteria interaction, plant-mycorrhiza interaction	RB1-29 TB-22	Understanding the process of symbiotic relationship between beneficial microbe and plants.
5. Socio-microbiology	L 17 – 20: Social behavior of microorganisms, Microbial communications: Quorum-sensing, types of quorum-sensing, role of quorum-sensing in microbial interactions with host and environment, prospective Application of quorum-sensing mechanisms in medicine, sociobiology of biofilms	Reviews	Understanding the processes how bacteria communicate and show community behaviour
6. Microbes and Human health (Pathogenesis and Therapeutic Microbiology)	L21 – 27: Microbial interactions with humans, host-parasite interaction, pathogenicity of microorganisms, Tools used to understand host-microbe interaction	TB-23, RB1-33, Reviews	Understanding the process of disease infestation in man
	L28 – 30: Antimicrobial chemotherapy, drug resistance, Bacteria and viruses in cancer therapy	RB1-34	Knowing the advanced method of treating diseases using bacteria or their products
7. Industrial Microbiology	L31 – 33: Biocatalysis and Biotransformation, Protein engineering, Bioprospecting of natural enzymes for industrial use, Primary and secondary metabolites, fermented foods, beverages, Single cell protein, Probiotics	RB2-8, RB2-11,	Understanding the Application of microbes in various industries.
	L34 – 35: Application of Metabolic engineering in fuel generation, Bioenergy production using microbes	RB2-13, Reviews	Identifying the role of microbe or their products in sustainable fuel and energy
	L36 - 37: Biosensors	Reviews	Use of microbial processes as biosensors in healthcare.
8. Synthetic Microbiology	L38- 40 History of Synthetic microbiology, Various Tiers in Synthetic microbiology, Metabolic engineering, etc.	Reviews, class notes	Acquaintance with genome engineering methods and synthetic life.
	L40 – 41: Application of synthetic microbiology, Design of microbial circuits	Reviews, class notes	Understanding how to utilize the synthetic biology methods



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

List of Experiments:

1. Isolation of bacteria for diversity study
2. Molecular identification of bacteria and phylogenetic analysis
3. DNA fingerprinting by ERIC-PCR
4. Test of plant growth-promoting properties
5. Assays for biofilm formation by *E. coli* K12 strain.
6. Gentamycin protection assay + imaging
7. Estimation of MIC and MBC.

6. Evaluation Scheme:

EC No.	Evaluation Component	Duration	% weightage	Date, Time & Venue	Remarks
1.	Mid-Sem	90 min	30	05/10 AN2	CB
2.	Quiz/Assignments*		15		CB/OB
3.	Lab		15		
4.	Comprehensive	3 hours	40	03/12/ AN	Partly OB

*To be evaluated periodically

Chamber consultation hour: To be announced in the class.

Notices: All notices will be displayed on the Nalanda.

Make-up policy: Make-up decisions will be made on a case-by-case basis and only for genuine cases as determined by the team and validated by the appropriate body. No make-up for Lab component and Quizzes.

Instructor-in-charge
BIO G523